

Introduction to Information Retrieval

Information Retrieval
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The screenshot shows a train journey selection interface. At the top, it states: "The journey you are searching for: **Glasgow Queen Street (GLQ) to Edinburgh (EDB)**". Below this, it specifies "Single Thursday 15 January 2026 • 1 Adult • No Railcard". There are buttons for "Edit journey" and "Set journey alert".

The main section is titled "Please select a journey" for "Thursday 15 January 2026". It shows two train options:

Train	Departure	Arrival	Class	Price	Action
12:42 On time Glasgow Queen Street (GLQ) Platform 9	→	13:59 Edinburgh (EDB)	Single from	£16.60	☐ Select
1h 17m, Direct					
12:45 On time Glasgow Queen Street (GLQ)	→	13:36 Edinburgh (EDB) Platform 12	Single from	£16.60	☐ Select
51m, Direct					

At the bottom right, there is a "Total" section showing "£0" and a "Continue" button with the text "Review tickets and options".

Overlaid on the right side of the screenshot is a purple rounded rectangle containing the following text:

- Some Characteristics**
- Structured Query
- Structured Data
- Accuracy verified
- Useful Data is returned
- Exact match
- Answer meets query criteria

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Information Retrieval

Information Retrieval is the *science of search engines*



How best to address the **information needs** of users...

- *Effectively*: Get the right information to a user!
- *Efficiently*: Get it to users quickly!

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Information Needs



I want to know what buildings to see in Glasgow

~~Glasgow~~ ~~What~~ ~~stand~~ ~~ing~~ ~~buildings~~

Glasgow Architecture - A Walk About Town - Scotcities.com
www.scotcities.com/central.htm ▼
A look at the fascinating architecture of the landmark buildings in Glasgow's city centre.

Architecture in Glasgow - Wikipedia, the free encyclopedia
https://en.wikipedia.org/wiki/Architecture_in_Glasgow ▼
As a result, Glasgow has an impressive heritage of Victorian architecture: the Glasgow City Chambers; the main building of the University of Glasgow, designed ...
[Glasgow Style](#) - [Victorian era](#) - [Modern era](#) - [References](#)

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Search Engines

- The most visible application of information retrieval technologies are search engines
 - Search engines have evolved since their initial conception 70 years ago



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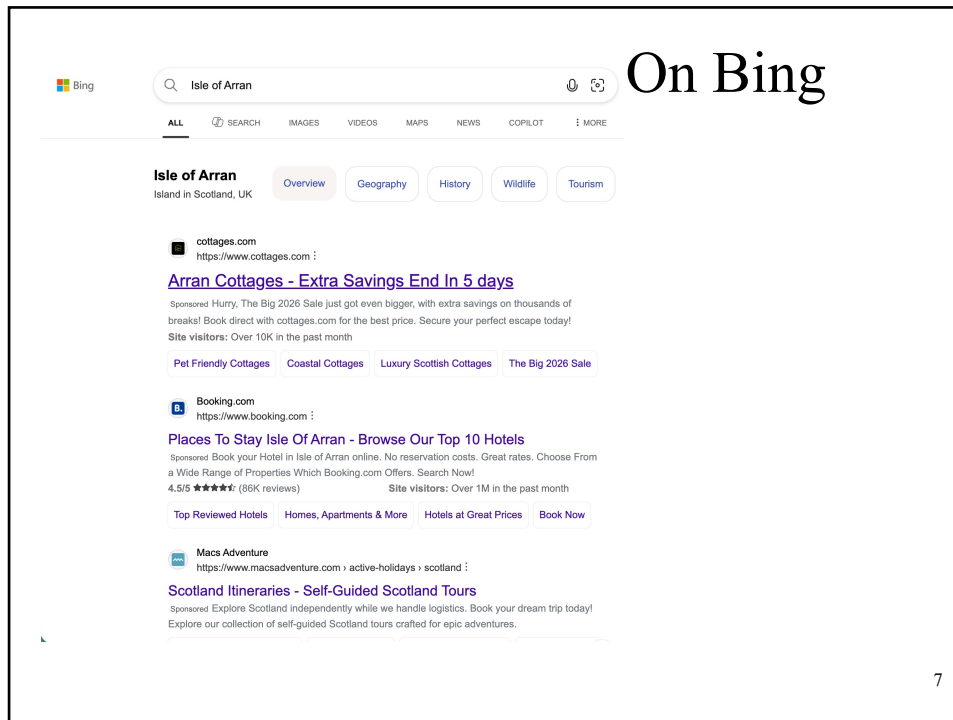
Search Engines



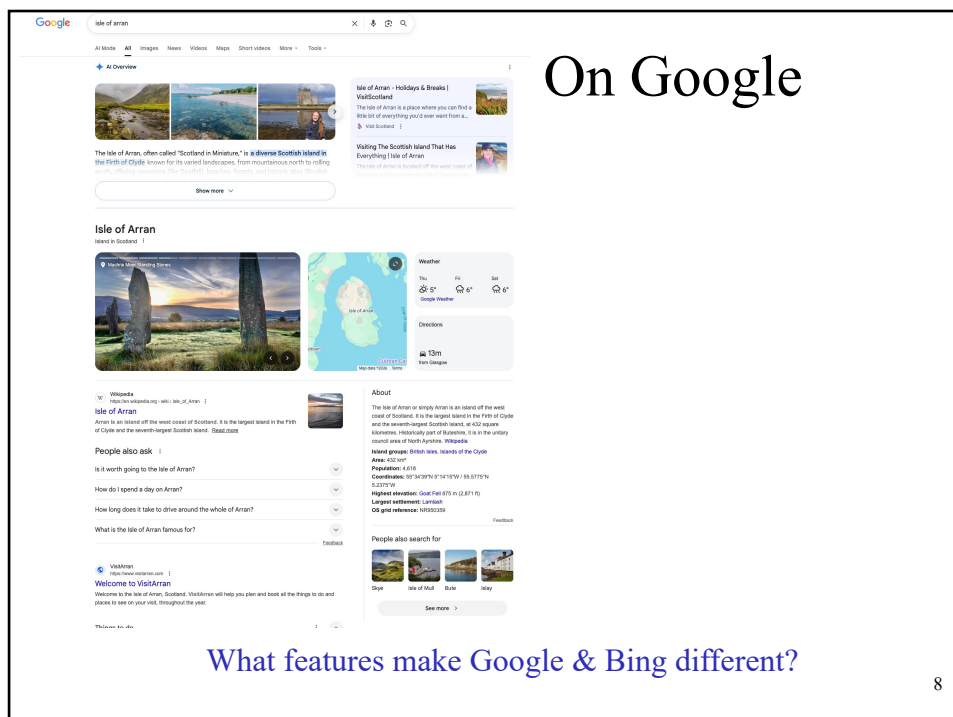
- Quite effective (at some things)
- Highly visible (some are very widely used)
- Commercially successful (some of them)
 - Google is one of the biggest corporations in the world
 - Underlying technology for searching
- What goes behind the scenes?
 - How do they work?
- Let us have a look at the commercial Search Systems!

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How do these systems work?

What are the commonalities & their functions?

Is there more to IR than Web Search?

The image shows four smartphones displaying different Google search results. The first phone shows a comparison table for 'butter vs olive oil' with nutritional information. The second phone shows restaurant recommendations for 'show me restaurants near my hotel'. The third phone shows a hotel reservation page for 'reserve a table there for tomorrow at?'. The fourth phone shows a list of bars for 'show me some bars around here'.

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What's the role of IR in AI Assistants?

The renaissance of IR in the ongoing AI era?

See Retrieval-Augmented Generation (RAG) ~ Large Language Models + IR

The image shows two screenshots. The left screenshot is the Microsoft Bing AI interface, featuring a 'Welcome to the new Bing' message and a search bar. The right screenshot is a detailed explanation of Information Retrieval (IR) and its role in AI assistants, including a definition of IR and a list of arguments for its importance.

What is information retrieval?

Information retrieval (IR) is the activity of obtaining material that can usually be documented on an unstructured nature i.e. usually text which satisfies an information need from within large collections which is stored on computers. IR systems are used to reduce what has been called information overload. The IR system assists the users in finding the information they require but it does not explicitly return the answers to the question. It notifies regarding the existence and location of documents that might contain the required information.

Would you like me to help you with anything else?

Arguments for IR being part of AI:

- **AI Techniques Employed:** Modern IR systems rely heavily on AI methods like machine learning, natural language processing, and deep learning. These techniques help understand documents, analyze user queries, and improve relevance ranking.
- **Interdependency:** Advancements in AI contribute significantly to advancements in IR and vice versa. For example, improved natural language processing leads to more accurate query understanding in IR systems.
- **Shared Goals:** Both AI and IR strive to mimic human intelligence, particularly in handling and making sense of information.

Arguments for IR being distinct from AI:

- **Focus on Retrieval:** IR primarily focuses on efficiently retrieving relevant information based on user queries. While AI encompasses a broader range of goals, including reasoning, planning, and decision-making.

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In This Course, We Ask ...

- What makes a system like Google or Bing tick?
 - How does it gather information?
 - What tricks does it use?
 - Expanding beyond the Web?
- How can those approaches be made better?
 - Natural language understanding?
 - Machine learning?
 - User interactions?
- What can we do to make things work quickly?
 - Fast computers? Caching?
 - Compression?
- How do we decide whether it works well?
 - For all queries? For special types of queries?
 - On every collection of information?
- What else can we do with the same approach?
 - Other media? Other tasks?

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Definitions of Information Retrieval

- Salton, 1968
 - Information retrieval is a field concerned with the **structure, analysis, organization, storage, searching and retrieval of information**
- Needham, 1977
 - The **complexity** arises from the **impossibility of describing the content of a document**, or the intent of **request**, precisely, or **unambiguously**
- **General definition**
 - Retrieval of **relevant information** from data sources which were not originally intended for access (e.g. unstructured data)
 - What does this mean?
 - Text (most often - e.g. searching newspaper articles or searching the Web)
 - Images... Video ... Audio....

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Documents vs. Database Records

- Database records (or **tuples** in relational databases) are typically made up of well-defined fields (or **attributes**)
 - e.g. bank records with account numbers, balances, names, addresses, social security numbers, date of birth, etc.
- Easy to compare fields with well-defined semantics to a query in order to find matches

(Unstructured) text is more difficult

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Imprecision in IR

- Most algorithms in Computer Science have a “**right**” answer. In contrast, a **heuristic** tries to guess something close to the right answer.
- Consider the three problems:
 - Sort the following ten integers
 - Find the highest integer
 - Find the beers made by X (i.e. SELECT ... FROM ...WHERE ...)
- Now consider:
 - Find the documents most **relevant** to “hippos in the zoo”.

IR techniques are essentially heuristics because we do not know the right answer

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What Makes a Document Relevant?

- It contains the query terms?
- It contains **ALL** of the query terms?
- It contains the query terms **many times**? (...but what if it is a long document?)
- It contains the query terms many times in a short document?
- It contains the query terms **close** together?
- It is **fresh/recent**?
- It contains terms **similar** to the query terms?
- It is **authoritative** (has many links)?
- It doesn't contain too many ads?
- It doesn't contain too many different, unrelated words (e.g. **spam**)
- It has been **clicked** by many other people for the same query?

ALL of these are heuristics. They are not guaranteed to get a correct, relevant document for all users

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Why to Retrieve?

- Task type ?**
- I need to find some information
 - Who is the head of college of science & engineering?
 - How to get to the LUX city centre from LUX airport?
 - *What is the upcoming topic in IR research?*
 - *What to do this weekend in Glasgow?*
 - Where to search?
 - Newspaper articles, Web pages, Scholarly materials (ACM/IEEE Digital Library), Emails, Tweets, ..., Images (Flickr), ..., Videos (YouTube)
 - ... and many more? Including a combination of all in your own desktop, ... or in your enterprise.... or external sources like the Web!

Time available varies

Exact Need

Vague need

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What do We Mean by “Information”?

- How is it different from “Data”?
 - Information is data in context
 - Databases contain data and produce information
 - IR systems contain and provide information
- How is it different from “Knowledge”?
 - Knowledge is a basis for making decisions
 - Many “knowledge bases” contain decision rules



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What Do We Mean by “Retrieval”?

- Find something that you are looking for:
 - **Ad hoc** search
 - Find documents “about this” topic-x
 - **Known item** search
 - Find the University of Glasgow home page
 - **Answer** seeking
 - What is the capital of Belgium?
 - Directed **exploration**
 - Who makes video conferencing systems?
 - **Decision** making
 - Best places to stay in Paris
 - **Expert** search
 - Who knows about Stable Marriage in my organisation?
 - Etc.

Role of Interaction

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Scenarios & Applications

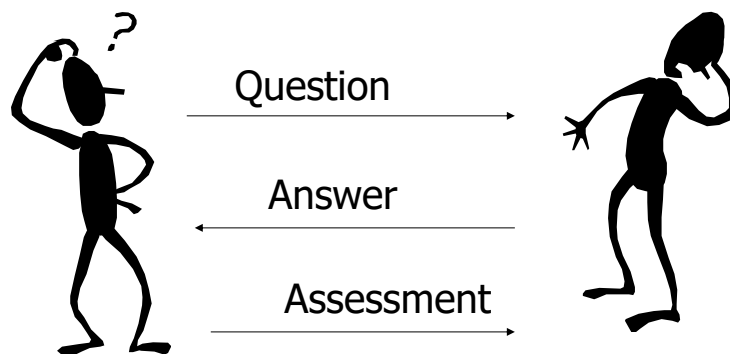
- Web search was a “killer app”
 - Developing advertising models on the web
- Today there are many retrieval applications:



- These types of content share common properties:
 - Text content + some metadata (e.g. title, author, date for papers; subject, sender, destination for email)

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A Question-answer scenario



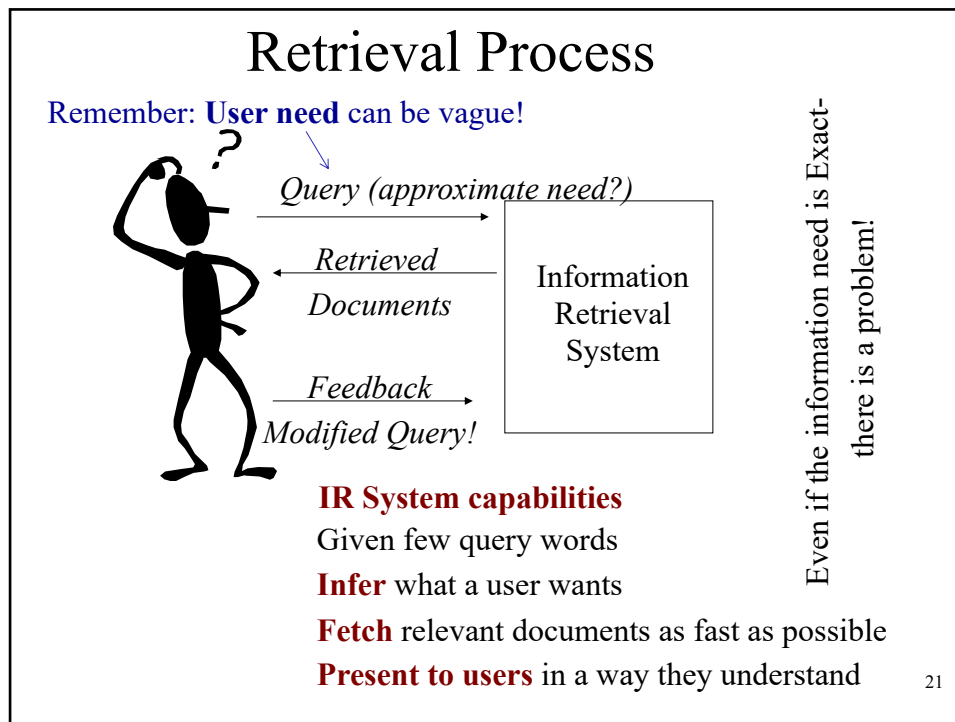
Note 1: Asking a good question can be as hard as answering it

Note 2: The objective of the **search engineer** is to automate the above process

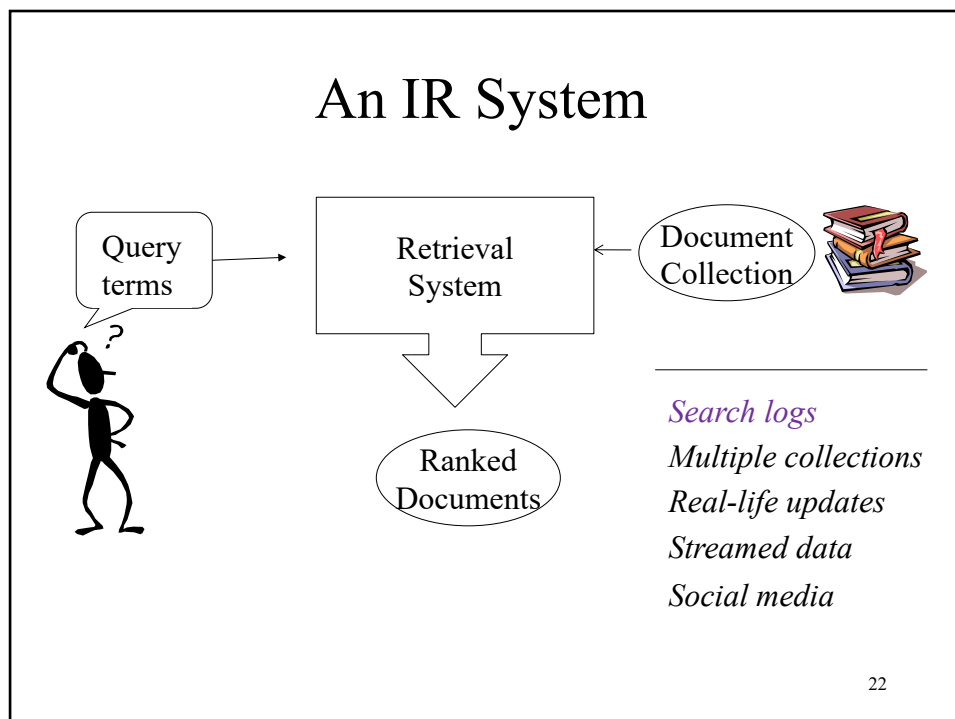
Search/IR Engineers are increasingly sought in industry

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Relevance ... Effectiveness ... Efficiency

- Relevance
 - If the query and the document are **about** the same topic (**known as the Topical Relevance**) – Remember that a user's query can be vague!!
- Effectiveness
 - Looks into the **quality** of the retrieved set. Do they largely contain **relevant** documents? (**role of Retrieval Models**)
- Efficiency
 - Return results as **fast** as possible (**role of search engine architecture - indexes**) – architecture of the IR systems: distributing computation, updating indexes, etc.

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IR H/M: General Information

- Structure
 - In-person lectures: Fridays 10:00-12:00 & [14:00-15:00]
 - Labs: BO1028 (cohort split into 2 groups) on Fridays 14:00-15:00 & 15:00-16:00 - Schedule of lab sessions will be announced on Moodle
- Assessment
 - Assessed coursework (25%), In-class quizzes (5%) & Final exams (80%)
 - Labs: Lab1 (Unassessed); Ex1 (13%), Ex2 (12%),
- Lecturer (s):

Craig Macdonald (Email: craig.macdonald@glasgow.ac.uk)

Sean MacAvaney (Email: sean.macavaney@glasgow.ac.uk)



Craig



Sean



@craig_macdonald
@macavaney

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Objectives

- At the end of this course you will be able to...
 - Explain the process of retrieval
 - Build an IR system and deploy it for practical applications
 - You may need a retrieval component in your own project work
 - Explain how IR systems are evaluated
 - Understand the web search engine architecture
 - Explain advanced IR technologies & Applications
 - Learning to rank; dense retrieval/neural IR; RAG, personalisation,
- Empower you with a necessary set of skills
 - Develop and deploy IR systems or related technologies

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Planned Syllabus

Fundamental Topics

Wk 1 - Starting on Monday 12th January - Introduction and IR Systems Architecture

Wk 2 - Starting on Monday 19th January - Term Weighting Models (Vector Space)

Wk 3 - Starting on Monday 26th January - Evaluation of IR Systems (Part 1); PyTerrier IR platform

Wk 4 - Starting on Monday 2nd February - Evaluation (Part 2), Relevance Feedback

Wk 5 - Starting on Monday 9th February - Advanced Retrieval Models (Probabilistic; Language Models; DFR, etc.)

Advanced/Emerging Topics

Wk 6 - Starting on Monday 16th February - Advanced IR Models; Machine Learning for IR & Learning to Rank; Web Search (introduction)

Wk 7 - Starting on Monday 23rd February - Web Search (introduction); Web Search (Link Analysis, Ranking);

Wk 8 - Starting on Monday 2nd March - Neural Networks for IR, PyTerrier hands-on tutorial on Neural IR

Wk 9 - Starting on Monday 9th March - Efficient IR (Compression, Pruning, DAAT/TAAT)

Wk 10 - Starting on Monday 16th March - Web Search Evaluation

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Coursework

- We will build upon open source libraries
- Deploying, developing and evaluating various retrieval techniques on a medium size Web collection
 - PyTerrier, Pandas, Gensim, Google Colab
- Expose students to a real large-scale IR system
 - Connect concepts from the lectures into code

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Tentative Schedule

Introductory Exercise

	Weight	Handout	Target for Completion
Lab 1	0%	w/c 19th January	Thu 5th February (suggested)

Assessed Exercises

Assessment	Weight	Handout	Due
Exercise 1	13%	w/c 2nd February	Wed 25th February, 4:30pm
Exercise 2	12%	w/c 22nd February	Thu 19th March, 4:30pm

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Feedback

- Feedback in this course take place in the following forms:
 - 1-1 discussion with lecturers and GTAs in labs
 - Discussions on Moodle
 - Personalised and general feedback on assessed coursework
 - Office Hours will also be available: c.f. Moodle

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You said, we did

- “Increase weight of coursework”
 - Adjusted exercise weighting (now worth 25%)
 - Added in-class quizzes (worth 5%)

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Teaching Resources

- Text Books – Recommended (available online!)
 - [Search Engines \(Information Retrieval in Practice\)](#), Bruce Croft, Metzer, Strohman, Addison Wesley 2015
 - [Introduction to Information Retrieval](#), Manning, Raghavan, Schutze (eds), Cambridge University Press 2008
- [Course Web page](#) is available on Moodle
 - Updated regularly as we go
 - News/material regarding the course will be posted on Moodle; Interactive quizzes
- Relevant Seminars
 - Mondays, 3-4pm; All are welcome
 - Announced on the school web pages

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First 3 Weeks: Look at the Basics of IR

- Architecture of the System
- Concepts of Relevance & Ranking
- Text Normalisation techniques
- IR Evaluation
- Conducting an IR experiment with PyTerrier

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